

CS6450 : Visual Computing

DYNAPROMPT

Dynamic Test-Time Prompt Tuning

ICLR 2025

Paper by:

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Roll No: **CS25MTECH14014**

Core Challenge

- ▶ Pretrained vision-language models (**CLIP**) degrade under **distribution shift** at test time
- ▶ No labeled data is available in the target domain

Types of Shift

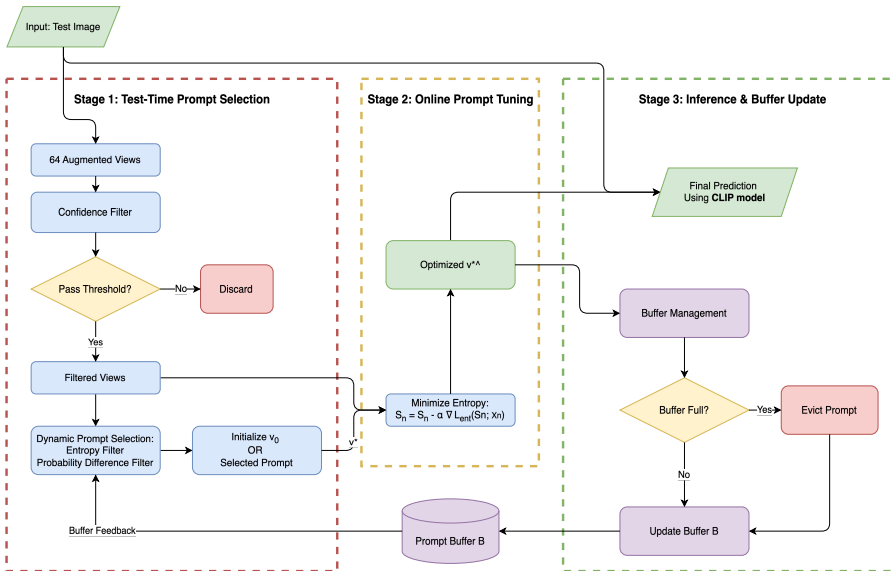
- ▶ **Domain shift:** real $\rightarrow$ sketch, corruptions, art styles
- ▶ **Cross-dataset:** ImageNet $\rightarrow$ satellite or medical data

Limitations of Existing Methods

- ▶ Require labeled data *or* ignore cross-sample knowledge
- ▶ Per-sample tuning (TPT) is stable but **does not reuse past information**
- ▶ Online tuning accumulates errors $\Rightarrow$ **prompt collapse**

Goal

- ▶ Adapt at test time without labels



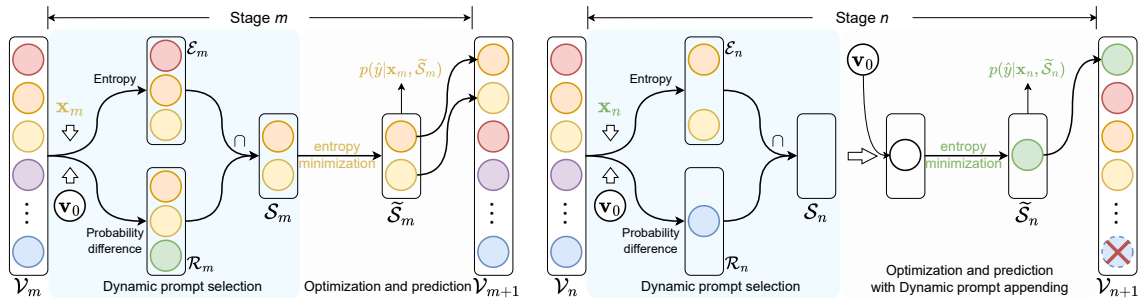


Figure. The process of our dynamic test-time prompt tuning. (Left) In dynamic prompt selection, we select the relevant online prompts from the prompt buffer $\mathcal{V}$ for each test sample using the intersection of the prompt subsets obtained by *entropy* and *probability difference* metrics. The selected prompts are optimized by entropy minimization before making predictions. (Right) If no prompt is selected, our dynamic prompt appending strategy assigns a new prompt initialized by $\mathbf{v}_0$ for the test sample and appends it to the prompt buffer. We always append the new optimized prompt on top of the buffer, moving the inactive ones to the bottom, which we can remove directly when appending new prompts to the full buffer.

Entropy-Based Selection

- ▶ **Calculation:**

$$\mathcal{D}_{\text{ent}}(\mathbf{x}_n, \mathbf{v}_i) = - \sum_{c=1}^C p_c \log p_c$$

- ▶ **Criterion** ($\mathcal{E}_n$):

$$\mathcal{E}_n = \{\mathbf{v}_i \in \mathcal{V}_n \mid \mathcal{D}_{\text{ent}}(\mathbf{x}_n, \mathbf{v}_i) \leq \mathcal{D}_{\text{ent}}(\mathbf{x}_n, \mathbf{v}_0)\}$$

- ▶ Retains prompt with **low entropy** than $\mathbf{v}_0$

Probability Difference Selection

- ▶ **Calculation:**

$$\mathcal{D}_{\text{pro}}(\mathbf{x}_n, \mathbf{v}_i) = p(c^* | \mathbf{x}_n, \mathbf{v}_i) - p(c^* | \mathbf{X}_n, \mathbf{v}_i)$$

- ▶ **Criterion** ($\mathcal{R}_n$):

$$\mathcal{R}_n = \{\mathbf{v}_i \in \mathcal{V}_n \mid \mathcal{D}_{\text{pro}}(\mathbf{x}_n, \mathbf{v}_i) \geq \mathcal{D}_{\text{pro}}(\mathbf{x}_n, \mathbf{v}_0)\}$$

- ▶ Retains prompts **sensitive** to augmentation

Combined: $\mathcal{S}_n = \mathcal{E}_n \cap \mathcal{R}_n$ **Prompts are confident and sensitive**

Test-Time Optimization

$$\tilde{\mathcal{S}}_n = \mathcal{S}_n - \alpha \nabla \mathcal{L}_{\text{ent}}(\mathbf{x}_n, \mathcal{S}_n)$$

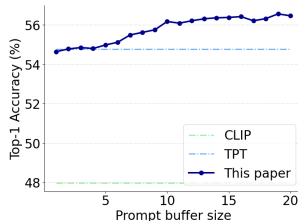
- ▶ Refine selected prompts via **entropy minimization** on $\mathbf{x}_n$
- ▶ Optimize prompt set $[[\mathbf{v}_1], [\mathbf{v}_2], \dots, [\mathbf{v}_k]]$

Impact of Buffer Size on Accuracy

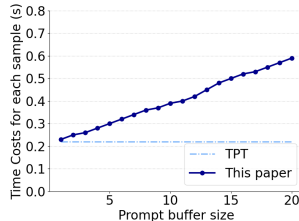
- ▶ Larger buffer retains more useful prompts $\Rightarrow$ better accuracy
- ▶ Larger buffer incurs more forward passes $\Rightarrow$ higher computational cost

Time Costs vs Buffer Size

- ▶ Cost grows linearly with M is $\mathcal{O}(M)$
- ▶ Larger M means more forward passes per sample



Accuracy vs. buffer size M on ImageNet-A.

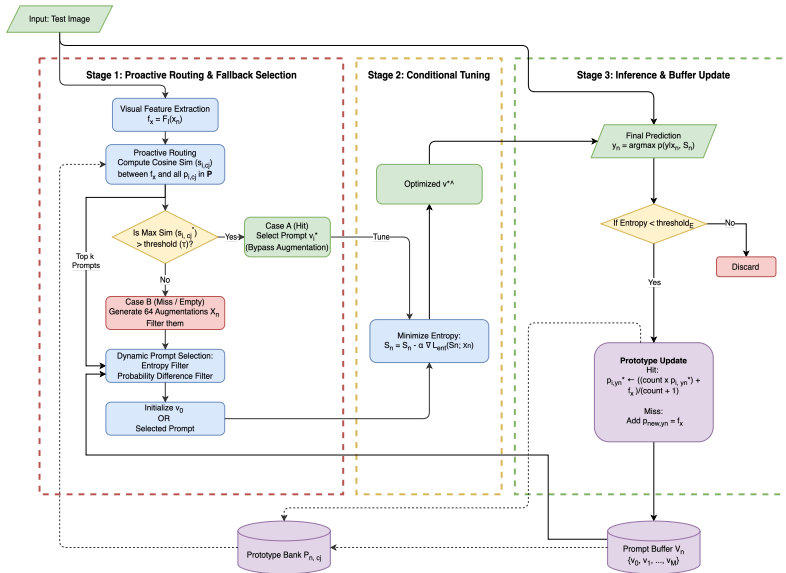


Per-sample inference time vs. buffer size

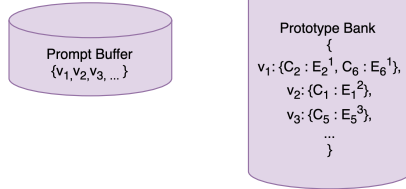
M .

- ▶ **Optimal Choice:** Best accuracy–cost trade-off at $M = 10$

- ▶ **Linear Computational Complexity:** The selection process requires evaluating all prompts in the buffer, resulting in a per-sample complexity of $\mathcal{O}(M)$.
- ▶ **Order Sensitivity in Online Adaptation:** The performance is dependent on the sequence of test samples, leading to variability under different data orderings.
- ▶ **Lack of Adaptive Buffer Mechanism:** The buffer size is fixed and does not adapt to changing data distributions, limiting robustness under distribution shifts.



n^{th} image



Embedding of the n^{th} image = E_n

Cosine similarity between E_c^v and E_n

	Class 1	Class 2	Class 3	Class 4	Class 5	Class 6
v_1		$\cos(E_2^1, E_n)$				$\cos(E_6^1, E_n)$
v_2	$\cos(E_1^2, E_n)$					
v_3					$\cos(E_5^3, E_n)$	

If the maximum cosine similarity is more than the threshold then it is cache hit.

Else cache miss

Find the maximum in each row and sort them in descending order. Pick the Top - k prompts for DynaPromptSelection.

► Proactive Latent Routing

- *Novelty*: Shifts from Selection-by-Evaluation to Selection-by-Association.
- *Reason*: Bypasses the $O(M)$ computational bottleneck of standard DynaPrompt by using image latents to directly map to the buffer, reducing inference latency by orders of magnitude.

► Unlimited Prompt Buffer

- *Novelty*: Continuous knowledge accumulation without information loss.
- *Reason*: Preserves specialized expert prompts for all encountered domains, preventing the **Catastrophic Forgetting** typically caused by fixed-size buffer eviction policies.

► Top- k Retrieval

- *Novelty*: **Constant-time complexity** in a growing memory space.
- *Reason*: Decouples computational cost from buffer size; regardless of how many thousands of prompts are stored, only the k most relevant are activated, ensuring real-time performance.

► Prototype Bank

- *Novelty*: Anchoring semantic knowledge to visual features.
- *Reason*: Acts as a **look-up table**. By storing the visual centroid alongside the prompt

► Entropy-based threshold

- *Novelty*: Autonomous buffer purification.
- *Reason*:
 - Prevent noisy, overfitted, or confused prompts from polluting the buffer.
 - Maintain good quality prompts

Backbone & Model

- ▶ Pretrained CLIP with **ViT-B/16**
- ▶ Prompt learning in **text embedding space**
- ▶ CLIP backbone **frozen**

Dataset & Evaluation

- ▶ Dataset: **Oxford Pets** (37 classes)
- ▶ Evaluation mode: **Test-time adaptation**
- ▶ Sample size: 200 images, 500 images.

Training Configuration

- ▶ Learning rate: **0.005**
- ▶ Batch size: **64**
- ▶ Context tokens: **4** (“a photo of a”)

Prompt & Buffer Settings

- ▶ Number of prompts: **5**
- ▶ Buffer size: **10**
- ▶ Top- k selection: **5**

Proposed Method Settings

- ▶ Proactive routing: **Enabled**
- ▶ Routing threshold: **0.85**
- ▶ Unlimited buffer: **Enabled**

Adaptation & Filtering

- ▶ Entropy threshold: **0.1**
- ▶ Fallback threshold: **0.4**
- ▶ TTA steps: **1**

Standard DynaPrompt vs Our Method

```
All prompts are used, reset prompt 0
[0, 4, 1, 3, 2]
Sample: 197, Time: 4.481s, Raw entropy: 1.23, Adapted entropy: 0.96, Raw pred: 23, Adapted pred: 9, Target: 23
All prompts are used, reset prompt 4
[4, 1, 3, 2, 0]
Sample: 198, Time: 4.491s, Raw entropy: 0.04, Adapted entropy: 0.00, Raw pred: 25, Adapted pred: 25, Target: 25
All prompts are used, reset prompt 1
[1, 3, 2, 0, 4]
Sample: 199, Time: 4.373s, Raw entropy: 0.95, Adapted entropy: 0.81, Raw pred: 16, Adapted pred: 16, Target: 35
Test: [199/200] Time 4.375 ( 4.161) Acc@1 0.00 ( 85.50) Acc@5 100.00 ( 96.50)
=> Total evaluation time: 832.18s
=> Acc. on testset [pets]: @1 85.5 / @5 96.5
```

```
Bypassing DynaPrompt Selection after False Hit.
Selection empty: discovering new domain and specializing expert.
Discovery: Added new prompt at index 32. Buffer size: 33
Test-time process completed
Running final inference
Sample 199: Criteria NOT met (Ent: 1.0126, Raw: 0.9282). Prompt dropped. Total dropped: 138
Test: [199/200] Time 6.508 ( 5.190) Acc@1 0.00 ( 88.00) Acc@5 100.00 ( 96.50)
* Acc@1 88.000 Acc@5 96.500
=> Total evaluation time: 1037.95s
Saved proactive routing stats to runs/exp_63/logs/routing_stats.json
Evaluation completed. Top1: 88.00, Top5: 96.50, Dropped: 138, False Hits: 35
=> Acc. on testset [pets]: @1 88.0 / @5 96.5
All datasets evaluated. Printing summary...
```

Observation

Accuracy improved from **85.5%** to **88%** (+2.5%) over 200 samples.

Standard DynaPrompt vs Our Method

```
All prompts are used, reset prompt 1
[1, 0, 3, 4, 2]
Sample: 497, Time: 4.939s, Raw entropy: 1.44, Adapted entropy: 1.16, Raw pred: 32, Adapted pred: 32, Target: 26
All prompts are used, reset prompt 0
[0, 3, 2, 4, 1]
Sample: 498, Time: 4.729s, Raw entropy: -0.00, Adapted entropy: 0.00, Raw pred: 0, Adapted pred: 14, Target: 14
All prompts are used, reset prompt 3
[3, 2, 4, 1, 0]
Sample: 499, Time: 4.708s, Raw entropy: -0.00, Adapted entropy: 0.07, Raw pred: 0, Adapted pred: 15, Target: 15
=> Total evaluation time: 2634.18s
=> Acc. on testset [pets]: @1 84.88630305175781 / @5 96.19999694824219
```

```
Proactive Routing Hit! Prompt index: 18, Similarity: 0.8580
Fine-tuning Hit prompt index 18 on current sample...
Hit confirmed. Post-adaptation entropy: 0.0136
Test-time process completed
Running final inference
Sample 499: Criteria NOT met (Ent: 0.0156, Raw: 0.0136). Prompt dropped. Total dropped: 398
* Acc@1 86.000 Acc@5 96.400
=> Total evaluation time: 2510.78s
Saved proactive routing stats to runs/exp_66/logs/routing_stats.json
Evaluation completed. Top1: 86.00, Top5: 96.40, Dropped: 398, False Hits: 96
=> Acc. on testset [pets]: @1 86.0 / @5 96.4000015258789
All datasets evaluated. Printing summary...
```

Observation

Accuracy improved from **84.8%** to **86%** (+1.2%) over 500 samples.

Key Research Outcomes:

- ▶ **Efficiency:** *Proactive Latent Routing* eliminates the $O(M)$ evaluation bottleneck, achieving constant-time $O(1)$ prompt selection.
- ▶ **Persistence:** An *Unlimited Buffer* **prevents Catastrophic Forgetting** by retaining specialized experts across all encountered domains.
- ▶ **Integrity:** *Entropy-based Quality Gating* filters noisy updates, ensuring the good quality prompts in the buffer.

Proposed Novelty	Technical Shift	Impact
Proactive Routing	Reactive $\rightarrow$ Predictive	$\downarrow$ Latency
Unlimited Buffer	Fixed $\rightarrow$ Cumulative	$\uparrow$ Robustness
Top- k Retrieval	Global $\rightarrow$ Localized Activation	Constant-Time Complexity
Entropy threshold	Unfiltered $\rightarrow$ Quality-Aware	$\uparrow$ Reliability

- ▶ **Adaptive Top- k Selection:** Implementing an uncertainty-driven retrieval mechanism that dynamically adjusts k based on real-time prediction confidence (Entropy-aware retrieval).

- ① Xiao Z., Yan S., Hong J., Cai J., Jiang X., Hu Y., Shen J., Wang Q., Snoek C. G. M. **DynaPrompt: Dynamic Test-Time Prompt Tuning.** *ICLR 2025*.
- ② Shu M., Nie W., Huang D.-A., Yu Z., Goldstein T., Anandkumar A., Xiao C. **Test-Time Prompt Tuning for Zero-Shot Generalization in Vision-Language Models.** *NeurIPS 2022*.
Learning to Prompt for Vision-Language Models (CoOp). *IJCV 2022*.
- ③ Khattak M. U., Rasheed H., Maaz M., Khan S., Khan F. S. H. **MaPLe: Multi-modal Prompt Learning.** *CVPR 2023*.

Thank You!

Questions?